Paper 1, Theorems 1.2–1.3: Statement Amendment

Internal research note for the `ShenWork` formalization

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Abstract

The stability proof in Section 5 of Shen's traveling-wave paper has six independent statement-level defects. The perturbed quadratic in (5.31) is positive at the unperturbed tail exponent, so the root comparison in (5.35) has the wrong direction; the weighted norm in (1.21) is written in laboratory rather than moving coordinates; and the exponential factor following (5.35) has the wrong sign. In addition, the chemotactic flux difference in (5.18) has two reversed signs. Finally, (5.27) loses one factor of b_1 in Young's inequality and (5.29)–(5.30) drop the $M^{2(\gamma-1)}$ factor from Lemma 5.3. This note gives the direct amendment supported by the corrected calculation, explains the corresponding change to the uniqueness theorem, and records the exact Lean certificates and remaining analytic boundary.

1 Source checked

The source examined is the original submission, not a repository paraphrase:

> Wenxian Shen, *Existence, uniqueness, stability, and monotonicity of traveling waves for repulsion/attraction chemotaxis models with logistic type source*, arXiv:2605.04401v1, 6 May 2026.

The archived local copy is `paper1.pdf`. Theorem 1.2 claims weighted stability for every

$$\kappa < \eta < \frac{1}{1+|\chi|^{1/6}},$$

where $\kappa = (c-\sqrt{c^2-4})/2$, and writes the weighted conclusion in laboratory coordinates.

2 The root comparison in (5.35)

The energy coefficient obtained in (5.31) is

$$q(\eta) = \eta^2 - (c-A)\eta + (1+B), \tag{1}$$

where, in the notation of (5.32)–(5.33),

$$A = |\chi|^{1-3\sigma} D', \qquad B = |\chi|^{1-3\sigma} D''.$$

The displayed formulas give $A \geq 0$ and, when $\chi \neq 0$, $B > 0$. The unperturbed tail exponent satisfies

$$\kappa^2 - c\kappa + 1 = 0.$$

Substitution into (1) therefore gives the exact identity

$$q(\kappa) = A\kappa + B > 0. \tag{2}$$

Consequently κ cannot lie between the two roots of q. Under the paper's speed inequality it lies strictly below the perturbed lower root:

$$\kappa < \kappa^{-}_{c,\chi,m,\alpha,\gamma}.$$

By continuity, (2) also produces weights strictly above κ for which $q(\eta) > 0$. Thus excluding only the endpoint $\eta = \kappa$ does not repair (5.35). The global energy estimate proves decay only above the perturbed lower root.

3 The coordinate mismatch in (1.21)

Section 5 changes to the moving coordinate $z = x - ct$ and estimates

$$E_{\text{move}}(t) = \int_{\mathbb{R}} e^{2\eta z} \left|u(t, z + ct) - U(z)\right|^2 \, dz. \tag{3}$$

The printed formula (1.21) instead uses

$$E_{\text{lab}}(t) = \int_{\mathbb{R}} e^{2\eta x} \left|u(t, x) - U(x - ct)\right|^2 \, dx. \tag{4}$$

The substitution $x = z + ct$ gives

$$E_{\text{lab}}(t) = e^{2\eta ct} E_{\text{move}}(t). \tag{5}$$

Hence decay of (3) does not imply decay of (4) without an additional rate stronger than $2\eta c$, which the proof does not supply. The natural correction is to use (3), equivalently the weight $e^{2\eta(x-ct)}$ in laboratory coordinates.

4 The sign after (5.35)

The paper defines the value $\lambda = q(\eta)$ to be negative and then writes an upper bound with $e^{-\lambda t}$, claiming that it tends to zero. For $\lambda < 0$, however, $-\lambda > 0$ and this factor grows. With the displayed definition the decaying factor is $e^{\lambda t}$. Equivalently, define the positive rate $\rho = -\lambda$ and write $e^{-\rho t}$.

5 The two chemotactic sign errors in (5.18)

The elliptic equation in (1.1) is

$$v_{xx} - v + u^{\gamma} = 0, \qquad \text{hence} \qquad v_{xx} = v - u^{\gamma}.$$

Put $w = u - U$ and $z = v - V$. The zero-order part of the difference of the two chemotactic flux derivatives is

$$(va_m - a_{m+\gamma})w + U^m z. \tag{6}$$

After multiplication by $-\chi$, the perturbation equation must therefore contain

$$-\chi(va_m - a_{m+\gamma} + b_2)w - \chi b_4 z. \tag{7}$$

Equation (5.18) instead prints $+a_{m+\gamma}$ inside the first parentheses and $+\chi b_4 z$; the same two reversals propagate into (5.19).

The conservative energy budget can nevertheless be repaired without a new asymptotic constant. For either sign of χ,

$$|va_m - a_{m+\gamma}| \le (2m+\gamma)M^{m+\gamma-1},$$

and the $b_4 z$ term is estimated by absolute value. The $(2m+\gamma)M^{m+\gamma-1}$ contribution is already present in (5.33).

6 The lost square in (5.27)

The J_1 cross term obeys the direct Young estimate

$$|\chi b_1 W_x W| \le \frac{1}{2}W_x^2 + \frac{1}{2}|\chi|^2 b_1^2 W^2.$$

Formula (5.27), propagated into (5.33), uses b_1 rather than b_1^2. There is no hypothesis making this replacement an upper bound; in particular, the negative-χ branch allows arbitrary $|\chi|$. If B_1 bounds $|b_1|$, the corrected constant contribution is $|\chi|^2 B_1^2/2$.

7 The missing resolver power in (5.29)–(5.30)

Lemma 5.3 gives

$$\int_{\mathbb{R}} V^2 \le \frac{\gamma^2 M^{2(\gamma-1)}}{(1-\eta)^2}\int_{\mathbb{R}} W^2, \qquad \int_{\mathbb{R}} V_x^2 \le \frac{\gamma^2 M^{2(\gamma-1)}}{1-\eta^2}\int_{\mathbb{R}} W^2.$$

The subsequent estimates retain γ^2 but omit $M^{2(\gamma-1)}$. This power is not identically one: the positive-χ asymptotic cap M_χ can exceed one. For $\chi \ne 0$, put

$$x = |\chi|^{1/6}, \qquad K = 1 + \gamma^2 M^{2(\gamma-1)}\frac{(1+x)^2}{x^2}.$$

Whenever $0 \le \eta < 1/(1+x)$, this K bounds both $1 + R_V(\eta)$ and $1 + R_{V_x}(\eta)$. The case $\chi = 0$ is separate and all chemotactic corrections vanish.

8 Recommended amended statement

Let B_i bound $|b_i|$ and use the corrected common resolver factor K above. The valid replacements for (5.32)–(5.33) are

$$A_\chi = |\chi| B_1 + \frac{|\chi| B_3}{2}K,$$
$$B_\chi = |\chi|\left((2m+\gamma)M^{m+\gamma-1} + B_2\right) + \frac{|\chi|^2 B_1^2}{2} + \frac{|\chi|(B_3+B_4)}{2}K.$$

Set

$$\Delta = (c - A_\chi)^2 - 4(1+B_\chi),$$
$$\kappa^- = \frac{(c-A_\chi) - \sqrt{\Delta}}{2},$$
$$\kappa^+ = \frac{(c-A_\chi) + \sqrt{\Delta}}{2}.$$

Choose the speed threshold so that $\Delta > 0$ and

$$\kappa^- < \frac{1}{1+|\chi|^{1/6}} < \kappa^+.$$

Recommended amendment to Theorem 1.2. Replace the condition $\kappa < \eta$ by

$$\kappa^-_{c,\chi,m,\alpha,\gamma} < \eta < \frac{1}{1+|\chi|^{1/6}},$$

and replace (1.21) by

$$\lim_{t\to\infty} \int_{\mathbb{R}} e^{2\eta z} \left|u(t,z+ct;u_0) - U^*(z)\right|^2 \, dz = 0.$$

Retain the uniform conclusion (1.22), subject to the compactness and far-left rigidity argument in Step 4. Write the energy decay as $e^{\lambda t}$ for $\lambda < 0$, or as $e^{-\rho t}$ for $\rho = -\lambda > 0$.

Theorem 1.3 uses Theorem 1.2 with the second wave as initial data. Its common tail exponent must therefore leave room for the corrected stability weight: the hypothesis should require $\kappa_1 > \kappa^-$, not merely $\kappa_1 > \kappa$. The waves constructed in Theorem 1.1 satisfy a family of tail estimates, so such a rate can be selected once κ^- lies below the construction's tail cap.

9 What would be needed to retain the original interval

Equation (2) obstructs only the *global coefficient* estimate used in (5.31); it is not a counterexample to stability itself. The perturbation coefficients may decay in the right tail. Retaining every $\eta > \kappa$ would require a new spatially localized coercivity or weighted spectral argument. No such argument appears in Section 5. It would be a new theorem, not a formalization of the written proof.

10 Lean certificates and current boundary

The repository contains zero-`sorry`, standard-axiom certificates. The algebraic root certificates in `ShenWork/Paper1/Theorem12RootObstruction.lean` are

- `paper531_kappa_not_between_perturbed_roots`;
- `paper531_kappa_lt_rootMinus`;
- `paper531_positive_inside_stated_weight_window`.

The coordinate certificate `laboratoryWeightedL2Energy_eq_exp_mul_coMoving`, together with a non-vacuous coordinate witness, is in `ShenWork/Paper1/Theorem12CoordinateAudit.lean`. The two sign certificates `paper531_printed_decay_factor_tendsto_atTop` and `paper531_corrected_decay_factor_tendsto_zero` are in `ShenWork/Paper1/Theorem12RootObstruction.lean`. The corrected flux identity `paper5ChemFluxDifference_expansion_corrected` and its uniform zero-order estimate `paper5CorrectedChemZeroCoefficient_abs_le` are in `ShenWork/Paper1/Theorem12MeanCoefficients.lean`. The corrected Young, resolver-cap, and scalar-budget certificates `paper5J1VariableDensity_le`, `paper5WeightedResolverFactors_le_cap`, and `paper5ResolvedW2Coefficient_le_corrected531` are in `ShenWork/Paper1/Theorem12WeightedEnergy.lean`.

`ShenWork/Paper1/Theorem12Corrected.lean` proves the scalar Gronwall and moving-frame localization wiring, but keeps the general-data whole-line PDE energy and compactness input explicit. The full amended stability theorem is therefore not yet claimed as proved. The corrected structural reduction for Theorem 1.3 is in `ShenWork/Paper1/Theorem13Corrected.lean`; its remaining analytic input is the same bounded-solution stability theorem.